

General revision

Based on

Unit three: modern physics

Unit four: modern electronics(semiconductors)

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Unit 3(modern physics)

A) What is meant by

1) Wien's Law

The wave length at maximum radiation intensity (λ_{max}) is inversely proportional to the temperature on Kelvin scale.

2) Blackbody

It is an ideal system that absorbs all radiation incidents on it. Then re-emit this radiation totally

3) The photo electric effect

It is a phenomenon emission of free electrons from metallic surface as a result of light radiation falling on it.

4) The Thermionic emission

Emission of free electrons from metallic surface as a result of heat energy falling on it

5) The surface potential barrier

It is the attractive force between the positive ions and free electrons on the metal which make the electrons cannot leave it.

6) The work function:

It is the minimum quantity of energy is needed to escape "liberate" an electron from a metal.

7) Compton effect:

When x-ray or γ - rays photons collide with a free electron, the photon frequency decreases and its direction changes, at same time the electron velocity increases and its direction changes.

8) X-rays

Electromagnetic rays of short wavelengths lie between ultraviolet and gamma rays in the invisible range

9) LASER

Light Amplification by Stimulated Emission of Radiation

10) Spontaneous Emission

The atom relaxes from an excited state to a lower state after lifetime interval is over emitting spontaneously the energy difference in the form of a photon without the effect of an external photon.

11) Stimulated Emission

External photon of energy $E_2 - E_1$ passes by excited atom in E_2 to relax it before the lifetime interval is over to emit the energy difference in the form of a photon

12) Monochromatic light (laser)

The color of visible light has its own characteristic wavelength.

13) Coherence

The emitted photons are in phase both at the same time and in the same direction

14) Collimated

The emitted photons are parallel and do not diverge, even at great distances

15) Population inversion

The number of atoms in the excited state exceeds the number of atoms in the lower state

16) Active medium

The effective material required to produce laser

17) Pumping process

Exciting some of active medium particles into high-energy state by using an external energy source

18) Meta – stable excited level

It is an excited energy level characterized by its lifetime is relatively long nearly 10^{-3} s

B) Give reasons for each of the following.

1) The classical physics cannot explain Planck's curve which shows the relation between the radiation intensity and the wave length

Because in classical physics $E=h\nu$ so the energy increase as the frequency increases but in plank's curve the intensity decreases at high frequencies

2) If we have a closed cavity with a small hole inside the cavity appears black

Because the most of radiation remain trapped due to successive reflections and small part which emits so the cavity appears black

3) In Compton effect the wavelength of the scattered photon is longer than that of the incident photon

Because the energy of incident photon is greater than that of the scattered photon due to the collision with a free electron (The photon gives part of its energy to a free electron, in the form of kinetic energy)

4) The wave length of the wave association the motion of electron decrease by increasing its velocity

According to **De Broglie equation** ($\lambda = h /mv$), the wavelength is inversely proportional to velocity.

5) Photons cannot be accelerated while electron can be accelerated

Because photons are electrically neutral particle and they are not affected by the either electric or magnetic field(moves with constant speed equals 3×10^8 m/s) while electron is a negatively charged particle affected by electric field and magnetic field.

1) Existence of the electric field (DC high voltage) between the cathode and the anode in generating X-rays in Coolidge tube

Because the electric field gives the emitted electrons from the filament high kinetic energy to collide with the target so its energy converted to X -rays. (i.e. the electrons will be accelerated)

2) X-rays are used to study the crystalline structure?

Because X-rays have ability to diffract, as they penetrate materials, the atoms in the crystal act as a diffraction grating so bright and dark fringes form, depending on the difference in the optical path

3) X-rays are used to detect defects in metallic structures

Because X-rays have high penetration power

4) X-rays are used to detect fractures

Because X-rays are used as imaging the bones and they affect sensitive photographic plates.

5) Tungsten commonly used as the target material in Coolidge tube

Because it has higher atomic number, the higher atomic number of target material, the shorter wavelength of the characteristic radiation will be

6) The anode in Coolidge tube made of copper that is fitted with fins

Because the copper characterized by high conductivity (low resistance, thicker) and fitted with fins to reduce the thermal energy due to collision

7) Laser is highly monochromatic in respect of the wavelength.

Because the spectral line has a very limited bandwidth (one wavelength)

8) Laser beam propagates in straight lines approximately parallel to each other.

Because the diameter stays constant for long distances without much scattering, thus energy is transmitted without much loss (the emitted photons are collimated)

9) Laser beam does not obey the inverse square law

That's mean its intensity remains constant over long distances and the emitted photons are coherent and propagate in one direction as a collimated parallel beam (the divergence is nil).

10) The active medium must be in population inversion state for laser sources to operate but that is not essential for ordinary light

Because it is necessary to have a large collection of atoms in the excited state for the laser to work efficiently

Population inversion means that the number of atoms in the excited state exceeds the number

11) Resonant cavity is an essential component in laser to complete the amplification and stimulated emission process

Resonant cavity is the container and activating catalyst for amplification which contains the active medium and has two parallel reflective mirrors on either side. The two mirrors return photons to the active medium, helping to increase the amount of energy that is produced

C) Mention the necessary conditions to obtain:

an electron from metallic surface when light incident on it	The energy of incident light is greater than the work function
Emission line spectrum	The atom relaxes from higher energy level to lower energy level
Characteristic radiation of X- rays	An electron emitted from cathode collides with an electron close to the nucleus of the target material atom
stimulated emission	photon of energy ($E_2 - E_1$) falls on an excited atom at level E_2 before life time is over (relaxation to lower level due to an external photon before lifetime is over)
Spontaneous emission	Excited atom relaxes from higher energy level to lower energy level to emit the energy difference in the form of a photon after lifetime interval is over
laser action	a) the active medium must be in population inversion b) stimulated emission c) amplification process

D) Mention one function or application only for each of the following.

Thermal imaging (remote sensing)	1) In criminology 2) Night vision system 3) Satellites and airborne which map and photograph the earth's surface. And the scientists analyze these images to determine the natural earth resources
Photoelectric cell	Convert the optical energy to electric energy
Cathode ray tube	In TV and computer monitors
Grid in cathode ray tube	Control the number of electrons in the electron beam
Anode in cathode ray tube	It is used to accelerate the electron beam towards the screen

	i.e. it Picks up the electrons which are liberated from the cathode, thus causing current in the external circuit
Electron gun	It is an instrument to produce a beam of electrons at high speed
X-rays	1) Studying the crystalline structure 2) Detecting the defects in metallic structure 3) They are used in imaging bones and fractures and some medical diagnosis
Coolidge tube	Production of X-rays
High dc voltage in Coolidge tube	Accelerate and direct the electrons emitted from the filament towards the anode to collide with the target element so the kinetic energy of emitted electrons converted to X -rays high electric field gives.
Cooling fins in Coolidge tube	Decreasing the heat generated due to collision between the emitted electrons from cathode and anode
Source of energy in laser tube	To excite the atoms of active medium
Resonant cavity	it is the container and the activating catalyst for amplification

E) Mention the factors affecting each of the following:

Energy of a moving photon (E)	Frequency of the photon ($E \propto \nu$) Or the wavelength of photon ($E \propto 1/\lambda$)
Kinetic energy of emitted electrons	The Frequency of the incident photon Or the wavelength of incident radiation
velocity of emitted electrons	The Frequency of the incident photon Or the wavelength of incident radiation
Work function	Kind of metal
Wavelength of characteristic radiation of X- rays	The kind of target element
The minimum Wavelength of continuous radiation of X- rays	The potential difference between the cathode and anode

F)Mention the scientific idea for each of the following

Cathode ray tube	Thermionic emission The electrons are produced from the cathode due to heat energy
Photoelectric cell	Photoelectric effect
Black body radiation	the particle nature of waves
Compton effect	the particle nature of waves
Night vision system	Thermal imaging
Laser	1) driving the atoms or molecules of the active medium into a state of Population inversion 2) stimulated emission

G) Compare between each pair

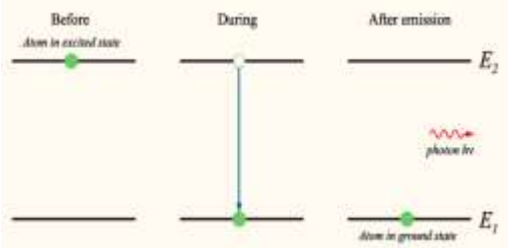
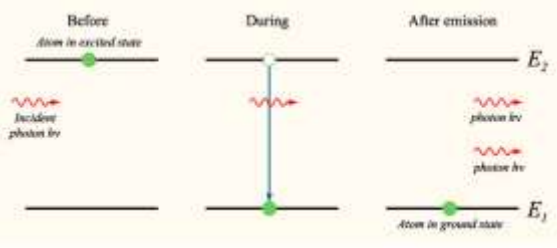
1) Characteristic radiation and Continuous radiation of X-rays

Characteristic radiation	Continuous radiation
The line spectrum is generated when an electron collides with an electron close to the nucleus of the target material atom. The electron in an atom of the target metal excited to a higher energy level because it gains sufficient energy and is replaced by another electron from a higher level to fill the space in the lower energy level.	This radiation is generated when the velocity of the colliding electrons is reduced upon passing through a retarding electric field due to repulsion, collision, and scattering. Hence, an electromagnetic radiation is generated based on Maxwell – Hertz theory
Depends on the target material atom	Depends on the potential difference between the cathode and anode
The difference in energy appears as radiation with a definite wavelength $\Delta E = \frac{hc}{\lambda}$	the shortest-wavelength radiation that can be produced is given by the following relation $eV = \frac{hc}{\lambda_{\min}}$
This radiation is called hard X-rays	This radiation is called soft X-rays

2) Ordinary light sources and Laser sources

Comparison	Ordinary light sources	Laser sources
Monochromatic	Each line in the visible spectrum includes a band of wavelengths.	A laser source emits one spectral line with a very limited band.
Collimation	The diameter of the emitted light beam increases with distance.	The diameter stays constant for long distances without scattering.
Coherence	Photons emit at different instants of time and varying phase.	Photons come out together at the same time and same phase difference.
Intensity	The light intensity of radiation falling on unit area decreases as the light source goes away due to spreading and is subject to inverse square law.	The laser rays falling on a unit surface are unspreading. They maintain a constant intensity and are not subject to the inverse square law.

3) Spontaneous Emission and Stimulated Emission

Point of comparison	Spontaneous Emission	Stimulated Emission
definition	The atom relaxes from an excited state to a lower state after lifetime interval is over emitting spontaneously the energy difference in the form of a photon without the effect of an external photon.	External photon of energy $E_2 - E_1$ passes by excited atom in E_2 to relax it before the lifetime interval is over to emit the energy difference in the form of a photon
drawing		
wavelength	The emitted photons have a wide range of wavelengths	The emitted photons are monochromatic (single) wavelength.
Propagation	The emitted photons propagate randomly	The emitted photons are coherent and propagate in one direction as a collimated parallel beam.
Intensity	The intensity of photons decreases according to the inverse square law. This is called spreading. While collisions with particles is called scattering. In ordinary light sources both spreading and scattering occur.	The intensity remains constant over long distances contrary to the inverse square law. It has been possible to send a laser beam to the Moon and receive it back, without much loss, despite the long distance involved. Spreading effect is nil and limited scattering takes place
example	This is the dominant radiation in ordinary light sources.	This is the dominant radiation in laser sources.

H) State the properties of each of the following

X-rays

- 1) They can penetrate media easily
- 2) They can ionize gases
- 3) They diffract in crystals
- 4) They affect sensitive photographic plates

Laser

- 1) Monochromaticity (spectral purity)
- 2) Collimation (photons are nearly parallel to each other)
- 3) Coherence (Photons are emitted at the same time, in phase and in the same direction)
- 4) intensity (it doesn't lose its intensity of illumination during propagation) (does not obey inverse square law)

Unit 4 (modern electronics)

A) What is meant by?

1) Dynamic equilibrium (thermal equilibrium)

The number of bonds broken per second will be equal to the number of bonds mended per second

OR

Fixed number of free electrons and free holes remains constant at every temperature

2) Forward Connection (forward bias)

The p-type of the diode is connected to the positive terminal and the n-type is connected to the negative terminal of a battery

3) Reverse Connection (Reverse bias)

The N-type is connected to the positive terminal and the P-type is connected to the negative terminal of the battery

4) Rectification

It is a process to convert an alternating current into a direct current by using a diode

5) Current gain (β_e)

It is the ratio between the collector current and the base current

6) Analog electronics

The electronics which deals with natural quantities

7) Digital electronics

A new branch of electronics has developed, such that the signal is in terms of one of two possible values representing two states 0 or 1 but is coded.

8) Logic Gates

These are the circuits that perform logic operations

B) Give reasons for each of the following:

1) At very low temperature, pure silicon crystal is an insulator

Because all bonds (covalent bonds) in the crystal are unbroken (there are no free electrons)

2) By adding some impurity atoms, the conductivity of semiconductors will increase

When a pure semiconductor is doped with impurity atoms (donor or acceptors), the charge carriers (electrons or holes) will increase, therefore the conductivity of semiconductors will increase due to movement the charge carriers

3) Emitter in a transistor is heavily doped

Because it emits charge carriers through the thin base layer to be collected by the collector

4) The thickness of the base of the bipolar junction transistor must be very small

Because very small quantity of electrons which emits from emitter is lost to recombination inside base, whereas, the rest of electrons get conducted through collector

5) Diode is used to a switch

Because the current flows in only one direction which is closed in the forward connection (conducting) and open (non-conducting) in the reverse connection

6) Digital electronics is preferred to analog electronics

Because the digital electronics have high ability to overcome the electrical noise and its basic idea is to code information in binary form (0 or 1)

7) The current gain(β_e) of a transistor is very large

Because it is the ratio between the collector current and the base current, the collector current is larger than the base current

C) Mention the necessary conditions to obtain:

P – type	The pure semi conductor is doped with trivalent atoms such as Boron
N – type	The pure semi conductor is doped with pentavalent atoms such as phosphors
Increasing the electrical conductivity of pure semiconductor	1) Increasing the temperature 2) Pure semiconductor is doped with impurity atoms (pentavalent or trivalent)

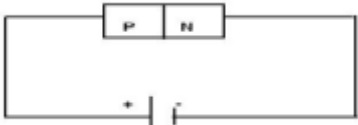
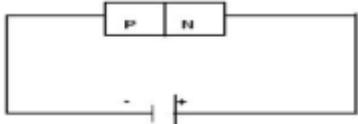
D)Mention one function or application for each of the following

the P-N junction (diode)	1) Rectification of ac current 2) switch
the transistor	1) Amplifier 2) switch
logic gates	1) Computer circuits 2) calculators, digital watches, computers, robots, industrial control systems, and in telecommunications

E)Mention the scientific idea for each of the following

Logic gates and digital electronics	digital electronics
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F)State the properties of each of the following:

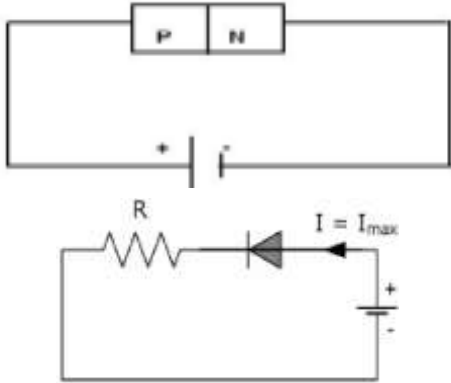
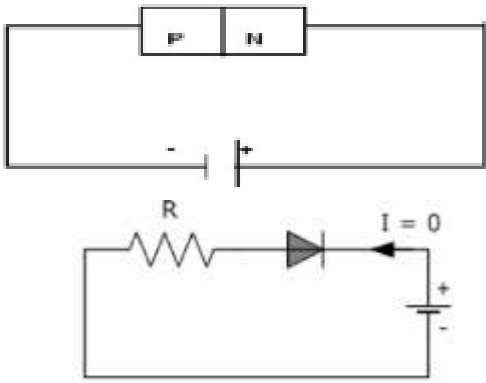
Semiconductors	a) They are materials with conductivities somewhere between conductors and insulators. Examples are silicon and germanium b) Pure semiconductors have low conductivity at low temperature. c) Pure semiconductors behave as prefect insulator at 0°K	
Forward connection	a) The depletion layer is narrow b) The resistance of diode decreases (Diode has a low resistance) c) Large current flows through the diode.	
Reverse connection	a) The depletion layer is wide b) The resistance of diode increases (Diode has a high resistance) c) very small current flows through the diode (can be neglected)	

H) Compare between each pair:

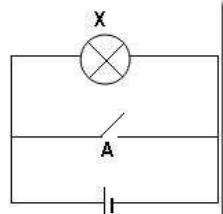
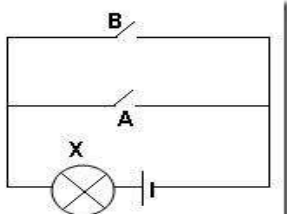
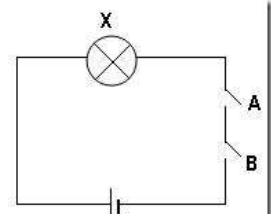
1) N-type and P-type

Point of comparison	N-type	P-type
Doping atoms (impurities)	Pentavalent (donor atom) (As or P or Sb)	Trivalent (acceptor atom) (Ga or Al or B)
Type of free charge (charge carriers) (majority)	Electrons(negative charges) ($n > p$) n is greater than p	holes(positive charges) ($p > n$) p is greater than n

2) Forward bias and reverse bias

Point of comparison	Forward bias	reverse bias
definition	The P-type of the diode is connected to the positive terminal and the N-type is connected to the negative terminal of a battery.	The N-type of the diode is connected to the positive terminal and the P-type is connected to the negative terminal of a battery.
Drawing		
The depletion region	The depletion layer is narrow, and the resistance of diode decreases (Diode has low resistance). Hence a large current flows through the diode	The depletion layer is wide, and the resistance of diode increases (Diode has a large resistance). Hence only a very small current flows through the diode (leakage current)

3) Different logic gates

Not gate	OR gate	AND gate																																				
only one input and one output	two inputs or more and has one output	two inputs or more and has one output																																				
a low-voltage input (0) is converted to a high voltage output (1)	Produces an output of 1(high) if either or both of the inputs are a 1	Produces an output of 1(high) when BOTH the inputs are a 1 (high), otherwise the output is 0 (low).																																				
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General revision based on unit three & four

Mention only the name of one apparatus whose function is based on each of the following and mention one use of it:

The scientific base	The apparatus	The use
thermal effect of black body radiation (thermal imaging)(remote sensing)	Night vision system	Used in military purposes which detect and image moving objects in the dark
Photoelectric effect (emission of electrons from metal due to light)	Photoelectric cell	Convert the optical energy into electric energy
Thermionic emission (emission of electrons from metal due to heat)	Cathode ray tube (CRT)	Used in TV and computer monitor

When does each of the following values become equal to zero?

1) The electrical conductivity of pure semiconductors (The number of free electrons in semiconductor crystal) At absolute zero (0°K)
2) The maximum kinetic energy ($k.E.$) _{max} of emitted electrons from the metal (velocity of emitted electrons from the metal) The frequency of incident light = the critical frequency Or the energy of incident light = the work function
3) Mass of photon At rest

How to increase each of the following:

1) Kinetic energy of emitted electrons from a metallic surface by using monochromatic light of shorter Wavelength (higher frequency)
2) Kinetic energy of emitted electrons from the filament in Coolidge tube By increasing the voltage between the cathode(the filament) and the anode (target element)
3) The penetrating power of X-rays spectrum By increasing the voltage between the cathode(the filament) and the anode (target element)
4) The intensity of X-rays By using a heating source of higher heat
5) The electrical conductivity of pure semiconductor a) by increasing the temperature b) by inserting some impurity atoms to pure semiconductor (doping)

What happens.....? (Explain each of them briefly)

What are the results produced in each of the following

when the voltage between the cathode and anode in cathode ray tube increases

Ans: the velocity of electrons emitted from cathode will increase

The reason : according to the relation $eV = K.E. = \frac{1}{2}mv^2$

As the voltage between the cathode and anode ,the velocity of emitted electrons will increase

when Falling of light ray with high intensity on a metal surface while its frequency is less than the critical frequency

Ans: the electrons will not be emitted from the metallic surface

The reason :because The condition of emission of electrons from metal is the frequency of incident light is greater than the critical frequency

when Falling of light ray with high intensity on a metal surface while its frequency is more than the critical frequency

Ans: the electrons will be emitted from the metallic surface

The reason :because The condition of emission of electrons from metal is the frequency of incident light is greater than the critical frequency

when the rate of photons (intensity) of incident light on a metallic surface is doubled while its frequency is more than the critical frequency(with respect to the rate of electron emission)

Ans:the rate of electron emission(the photocurrent)will be doubled

The reason :the rate of electron emission (the photocurrent) is directly proportional the intensity of light when the frequency of light is more than the critical frequency

when the intensity of a beam of monochromatic light is doubled while its frequency is more than the critical frequency (with respect to the momentum of each photon in the radiation)

Ans:the momentum of each photon in the radiation does not change

The reason: according to the relation $P = \frac{h}{\lambda}$

The wavelength is the same (monochromatic light) ,so the momentum of each photon does not change

A photon of gamma or (X) ray is incident on a free electron

Ans:the photon frequency(energy) (momentum) (mass) decreased and changed its direction; also the electrons velocity increased and changed its direction

The reason: the collision between the photon and electron is an elastic collision

1) The conservation of momentum

The total linear momentum before collision = the total linear momentum after collision

2) The conservation of energy

The sum of energy of photon and electron before collision = the sum of energy of photon and electron after collision

when the target in Coolidge tube is replaced by another metal (with respect to the continuous and characteristic spectrum of X rays)

Ans:the continuous spectrum of X rays does not change while the characteristic spectrum of X rays will change

The reason:the continuous radiation depends on the voltage difference between the cathode and the anode (the accelerating voltage)

the characteristic radiation (at the peak) does not depend on the voltage difference but depends on the target element

when the target in Coolidge tube is replaced by another metal of higher the atomic number (with respect to the wavelength of the characteristic spectrum of X rays)

Ans:the wavelength of the characteristic spectrum of X rays will decrease

The reason:The wavelength of the characteristic radiation depends on the target element

The higher the atomic number of the target element, the shorter the wavelength of the characteristic radiation

When a photon of energy $\Delta E = E_2 - E_1$ passes by an excited atom in the higher energy level E_2

Ans: stimulated emission takes place and the atom emits the excitation energy in the form of a photon of the same frequency, phase and direction of the falling photon.

The reason: the external photon stimulates excited atoms to emit the energy difference in the form of a photon before the lifetime interval is over

When pure Semiconductor crystal is cooled near 0°K

Ans :the electrical conductivity will be zero (the semiconductor will be a perfect insulator)

The reason:there are no free electrons due to all bonds in the crystal are unbroken

when the temperature of pure Semiconductor crystal increases

Ans :the electrical conductivity will be increased

The reason:the number of free electrons will increase due to some bonds in the crystal are broken

Some of the Silicon atoms in silicon crystal are replaced by some of phosphorus atoms

Ans :the electrical conductivity will be increased

the number of free electrons becomes more than that of holes

the crystal will be negative(N-type)

The reason: When a pure semiconductor is doped with Phosphorus (**P**) atoms donate free electrons to the crystal because the Phosphorus atom has 5 electrons, four of them will take part in the bonding and the excess electron will be freed it has more free electrons than holes

When connecting the P-N in electric circuit through reverse bias

Ans:very small current flows through the diode (is almost zero)

The reason :the depletion layer is wide and the resistance of diode increases (Diode has a high resistance)

When connecting the P-N in electric circuit through forward bias

Ans:high current flows through the diode

The reason :the depletion layer is narrow and the resistance of diode decreases (Diode has a low resistance)

Give reasons for each of the following:

A gamma ray photon makes a collision with an electron at rest; the momentum of photon will decrease

Ans: When a photon of high frequency (gamma rays) collides with a free electron, the photon frequency (energy or mass) decreased and Momentum of photon is given

$$P = mc = \frac{E}{c} = \frac{h\nu}{c}$$

the ordinary color appears to have different shades to the naked eye

Ans: Because the ordinary light includes a bandwidth of wavelengths

Questions and answers

Two stars, one of them appears as a blue and the other appears as a red, which the star has higher temperature? Why?

Ans.: The blue star has higher temperature

According to Wien's law ($\lambda_{\max} \propto \frac{1}{T}$)

$$\lambda_{\text{red}} > \lambda_{\text{blue}}$$

$$T_{\text{red}} < T_{\text{blue}}$$

Radiation of frequency 10^{15} Hz are incident on two photosensitive surfaces A and B.

Following observations are recorded:

Surface A: No photoemission takes place.

Surface B: photoemission takes place but photoelectrons have zero energy.

Explain the above observations on the basis of Einstein's photo electric equation

Ans.: surface A (the frequency of incident light is less than the critical frequency)

surface B (the frequency of incident light is equal to the critical frequency)

Why is the anode of Coolidge tube heated up at the time of emission of X-rays?

Ans: because there is a collision between fast moving electrons emitted from the cathode and the target element leading to heat up the anode

In cathode ray tube, what is the effect on the electron beam if

a) a hotter filament is used

If the temperature of the filament is high, then the number of electrons emitted per second will be more. As electrons are the beam particles so the beam particles will be more when a hotter filament is used

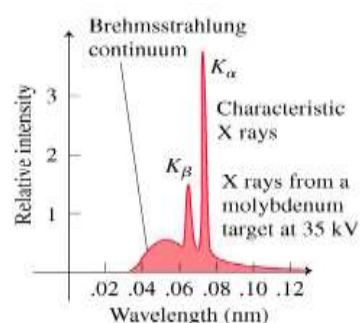
b) The anode voltage is increased?

When the anode voltage is increased the speed of the beam particles (electrons) will be more and hence the K.E of the beam particles will be increased

The opposite graph displays the spectrum of X rays emitted when electrons are accelerated through a potential difference of 35kv V and collided with a molybdenum target. The shortest wavelength X rays produced by this process are of about 0.035 nm.

a) If the accelerating voltage is decreased from 35kv to 30k V, determine what will happen to the wavelength of the shortest wavelength X rays produced

Ans: The wavelength will increase to greater than 0.035nm



b) If a larger atomic number element is used as a target, determine what will happen to the wavelength of the shortest wavelength X rays produced.

Ans: The wavelength will stay the same at 0.035 nm

c) If a larger atomic number element is used as a target, determine what will happen to the characteristic wavelength of the K_{α} line

Ans: It will decrease to a smaller characteristic wavelength

What is the basic difference between transistor as an amplifier and a step up transformer?

Ans:

The output power of an amplifier is greater than its input power, whereas the output power of a transformer is almost the same as its input power. In other words, an amplifier acts to increase power and voltage, whereas a transformer only increases voltage

If wavelength of electromagnetic waves are doubled what will happen to energy of photon

Ans: the energy of photon will be halved (there is an inverse relation between the energy and the wavelength)

Problems: Chapter5(wave particle duality)

Radiation of wavelength 400nm propagating in the form of parallel beam, fall normally on a metallic surface,. (given that $h = 6.6 \times 10^{-34}$ Js ,$c=3 \times 10^8$ m/sec) calculate 1) linear momentum 2) Energy of such radiation	$P = 6.6 \times 10^{-34} / 400 \times 10^{-9} = 1.65 \times 10^{-27} \text{ kg.m/s}$ $E = \frac{h c}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{400 \times 10^{-9}}$ $E = 4.95 \times 10^{-19} \text{ J}$
When electromagnetic radiation of wavelength 450 nm falls on the surface of metal, electrons are emitted with a kinetic energy of 1.68×10^{-19} J. What is the minimum energy needed to remove an electron from sodium? What is the maximum wavelength that will cause a photoelectron to be emitted? (given that $h = 6.6 \times 10^{-34}$ Js ,$c=3 \times 10^8$ m/sec)	$E = \frac{h c}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{450 \times 10^{-9}} E = 4.4 \times 10^{-19} \text{ J}$ $K.E. = E - E_w$ $E_w = 4.4 \times 10^{-19} - 1.68 \times 10^{-19} E_w = 2.72 \times 10^{-19} \text{ J}$ $E_w = \frac{h c}{\lambda_{max}}$ $2.72 \times 10^{-19} = \frac{(6.6)(10^{-34})(3)(10^8)}{\lambda_{max}}$ $\lambda_{max} = 7.28 \times 10^{-7} \text{ m}$
When monochromatic light of wavelength $5000 \text{ \AA}$ incident on a metal surface, electrons of velocity 2.57×10^5 m/s is released, is electrons release from this surface when other monochromatic light of wavelength $6000 \text{ \AA}$ incident on it? ? (given that $h = 6.6 \times 10^{-34}$ Js ,$c=3 \times 10^8$ m/sec)	$E_w = E - (K.E.)_{max}$ $E_w = \frac{h c}{\lambda} - \frac{1}{2} m_e v^2$ $E_w = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{5000 \times 10^{-10}} - \frac{1}{2} (9.1 \times 10^{-31})(2.57 \times 10^5)^2 = 3.6 \times 10^{-19} \text{ J}$ $E = \frac{h c}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{6000 \times 10^{-10}} = 3.3 \times 10^{-19} \text{ J}$

There is no emission of electrons when other monochromatic light of wavelength $6000\text{\AA}$ incident on it because $E < E_w$

(modern electronics)

Draw a transistor circuit as a switch at on condition then calculate the collector current of the transistor. A lamp having a resistance of $22\ \Omega$ is to be switched by a BJT. The power supply voltage (V_{cc}) is 5V and the collector emitter voltage is 0.1V. Calculate the minimum base current required to place the transistor in on condition if beta is 80

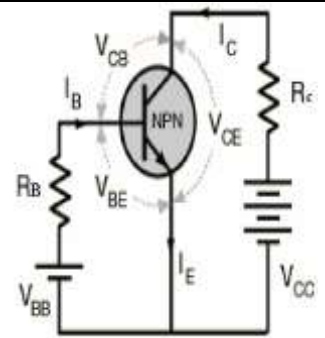
$$V_{cc} = I_c R_c + V_{ce}$$

$$5 = I_c (22) + 0.1$$

$$I_c = 0.22273\text{A}$$

$$I_c = \beta I_b$$

$$I_b = 0.00278\text{A}$$



In an NPN transistor 10^8 electrons enter the emitter in 10^{-8} s. If 1% electrons are lost in the base, calculate the current gain (the electron charge (e) = 1.6×10^{-19} C)

$N_e = I_e t$ **N: no. of electrons**

$$10^8 \times 1.6 \times 10^{-19} = I_e (10^{-8})$$

$$I_e = 1.6 \times 10^{-3}\text{A}$$

$$I_b = 0.01 \times 1.6 \times 10^{-3}\text{A}$$

$$I_c = 0.99 \times 1.6 \times 10^{-3}\text{A}$$

$$I_c = \beta I_b$$

$$0.99 \times 1.6 \times 10^{-3} = \beta \times 0.01 \times 1.6 \times 10^{-3}$$

$$\beta = 99$$

A transistor is connected in Common Emitter configuration. The voltage drop across the load resistance (R_C) $3\text{ k}\Omega$ is 6 V. Find the base current. α of the transistor is 0.97

$$V_c = I_c R_c$$

$$6 = I_c \times 3000$$

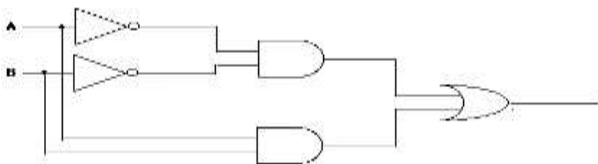
$$I_c = 0.002\text{A}$$

$$\beta = \frac{\alpha_e}{1 - \alpha_e} = \frac{0.97}{1 - 0.97} = 32.333$$

$$\beta = \frac{I_c}{I_b}$$

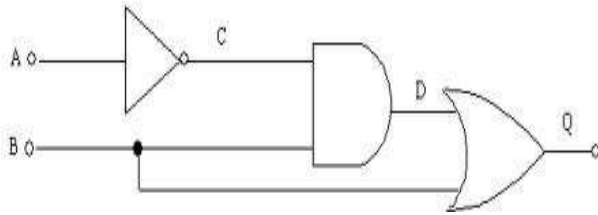
$$I_b = \frac{0.002}{32.333} = 6.185 \times 10^{-5}\text{A}$$

The electronic circuit below represents combining gates for a special job. Complete the following truth table for the following circuit



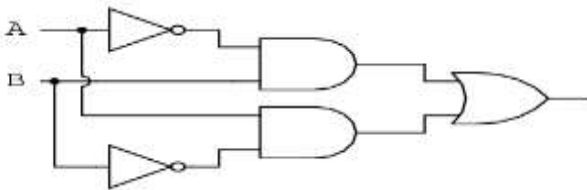
A	B	Output
0	0	1
0	1	0
1	0	0
1	1	1

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